IPCA
School of Technology

Report on the Practical Project for Machine Learning Algorithms Course

Nuno Miguel Carvalho Araújo, No. 20078

Lecturer: Luís Ferreira

Barcelos, 2026

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1 Introduction

1.1 Contextualization

This project, developed within the course unit of Machine Learning Algorithms, focuses on exploring good practices in the preparation, development, and application of intelligent models based on Machine Learning algorithms.

1.2 Motivation and Objectives

The aim of this project is to design and develop intelligent models capable of addressing real-world problems of moderate complexity in the domains of Classification or Regression. This involves identifying or creating appropriate datasets, performing rigorous data preparation, and applying effective Feature Engineering techniques. Following this preparation, one or more machine learning models will be selected based on well-defined success criteria, and their performance will be thoroughly evaluated. When necessary, improvement actions will be implemented to refine the chosen models. The use of pipelines will support automation and efficiency throughout the workflow, and model combination strategies will also be considered. Additionally, the project seeks to encourage the production of clean, well-documented source code and the development of a structured technical report.

The objectives of this project are as follows:

- Consolidate knowledge of Artificial Intelligence and Machine Learning algorithms.
- Analyze real-world problems and translate them into computational tasks.
- Develop programming skills in Python.
- Strengthen experience in designing and evaluating intelligent models.
- Deepen understanding of the content of the course unit.

Thus, this project aims not only to build and evaluate machine learning models for real-world scenarios, but also to promote a critical and well-founded understanding of the methodological, computational, and analytical processes involved, thereby consolidating essential skills in the field of Machine Learning.

2 Problem Definition and Scope

The problem under study consists of predicting short-term rental prices for Airbnb listings in Mexico City, Mexico. More specifically, the goal is to estimate the nightly rental price of a listing based on its observable characteristics, such as location, type of accommodation, and available amenities.

2.1 Market Context and Motivation

The short-term rental market in Mexico City has become increasingly competitive and dynamic, driven by the proliferation of platforms such as Airbnb. Prices fluctuate according to a variety of determinants, including neighborhood characteristics, accommodation type, and the specific amenities available. Accurately setting rental rates requires hosts to balance profitability with market competitiveness, while guests rely on clear and fair pricing for informed decision-making.

This study focuses exclusively on Mexico City, enabling an in-depth examination of local pricing patterns. By analyzing neighborhood-specific trends, accommodation distributions, and amenity offerings, the research aims to provide insights into the factors that influence pricing within this urban context.

The dataset employed originates from a publicly accessible Airbnb repository [1], offering structured details on listings across Mexico City, which serve as the basis for model training and evaluation.

2.2 Problem Formulation as Regression Task

The problem addressed in this project is formulated as a supervised machine learning regression task. Regression is appropriate because the target variable, namely the Airbnb rental price, is a continuous numerical value rather than a discrete class or category.

The model is trained using historical Airbnb listing data, where each instance consists of a set of input features (such as location, room type, and amenities) and a corresponding rental price. During training, the model learns a function that maps these input features to a real-valued output.

The expected outcome of the model is a numerical prediction representing the estimated rental price of an Airbnb listing given its characteristics. Model performance is assessed by comparing predicted prices with actual prices using standard regression evaluation metrics, with the objective of minimizing prediction error and achieving accurate and reliable price estimates.

3 Data Comprehension

This section presents a comprehensive exploration of the dataset, focusing on understanding its structure, statistical properties, and the relationships between features and the target variable. The analysis emphasizes the most relevant characteristics that influence rental pricing.

3.1 Dataset Overview

The dataset comprises 26,401 Airbnb listings in Mexico City with 79 features describing property characteristics, host information, location data, and pricing details. These features include 46 numeric and 33 categorical variables.

Initial examination revealed several data quality issues requiring preprocessing:

- The target variable `price` was stored as a string with currency symbols (e.g., "$1,200.00"), requiring conversion to numeric format.
- Multiple features contained significant proportions of missing values.
- The dataset included both numeric and categorical variables with varying scales and distributions.

As a preliminary step, the target feature `price` was converted to numeric format to facilitate subsequent analyses and data relationship exploration.

3.2 Target Variable Analysis

The target variable, nightly rental price, exhibits characteristics typical of real estate pricing data: right-skewed distribution with a long tail of high-value properties.

3.2.1 Descriptive Statistics

Table 1 summarizes the price distribution, revealing substantial variability across listings.

Table 1: Price Distribution Statistics

Statistic	**Value (USD)**
Mean	1,989.29
Median	1,039.00
Standard Deviation	18,500.68
Minimum	66.00
25th Percentile	630.00
50th Percentile	1,039.00
75th Percentile	1,647.00
Maximum	900,000.00

The substantial difference between mean and median indicates positive skewness, suggesting that while most listings have moderate prices, a subset of luxury properties significantly increases the average. This distribution pattern is consistent with urban short-term rental markets where premium properties command disproportionately higher rates.

Figures 1 and 2 present histogram representations of the rental prices.

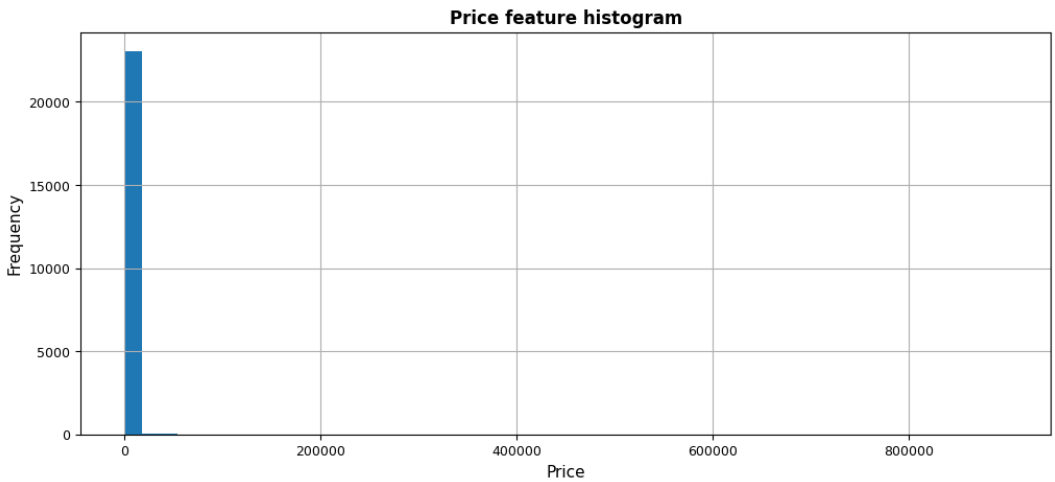

Figure 1: Complete price distribution.

Figure 1 displays the complete price range. However, extreme outliers extending beyond $10,000 per night severely compress the horizontal axis, obscuring the distribution of typical listings. To address this visualization limitation, Figure 2 excludes the top 1% of prices, revealing that most listings are concentrated below $5,000 per night with peak density in the lower price segments.

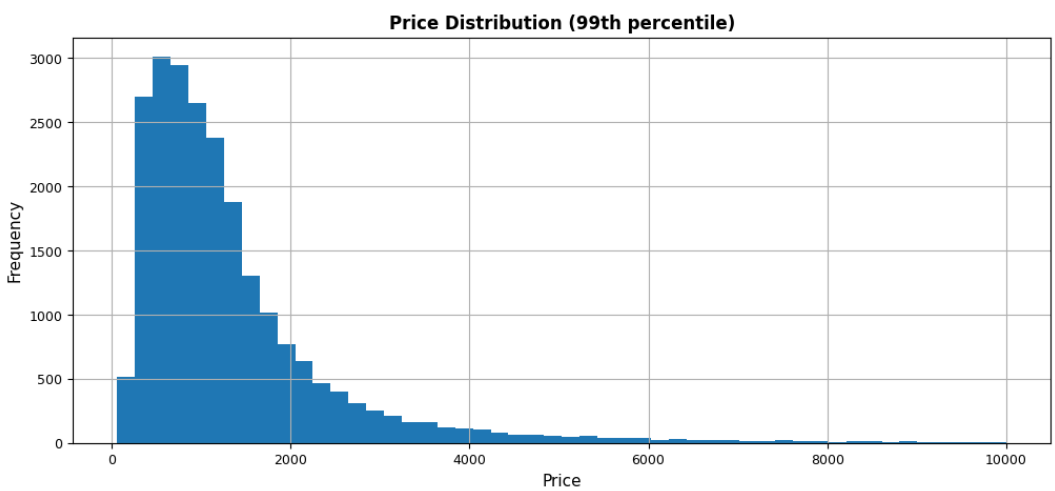

Figure 2: Price distribution up to the 99th percentile.

This right-skewed distribution confirms that extreme high-value properties constitute a small minority and indicates that outlier treatment will be necessary during data preparation.

3.3 Feature Relationships with Price

Understanding the relationship between property characteristics and rental prices is fundamental to effective feature engineering and model development.

3.3.1 Correlation Analysis

Initial correlation analysis across all numeric features revealed moderate linear relationships with price. Figure 3 presents the correlation matrix for numeric variables.

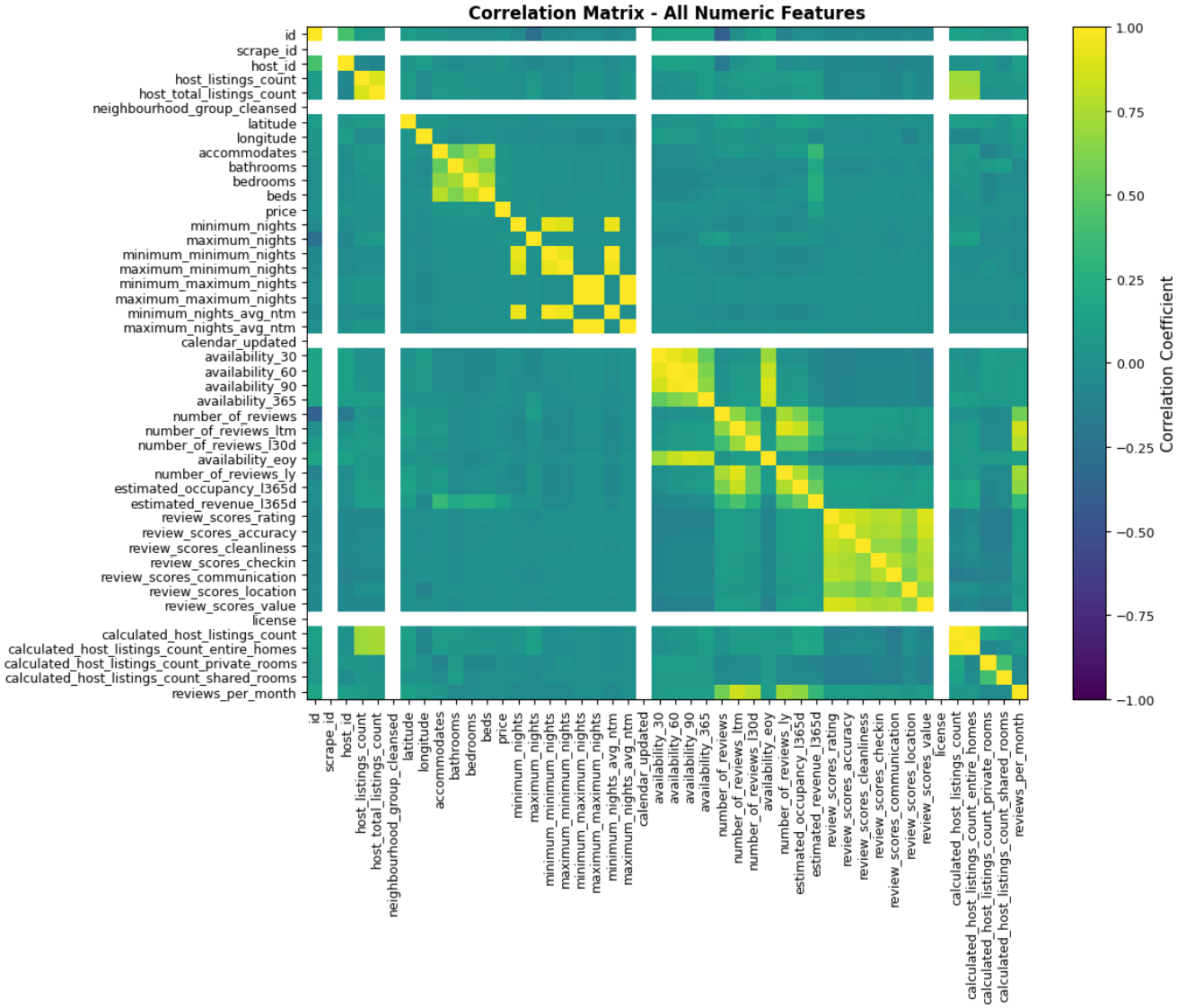

Figure 3: Correlation matrix heatmap - all numeric features.

While no single feature demonstrates exceptionally strong linear correlation with price, several property characteristics show meaningful associations worthy of further investigation.

3.3.2 Property Characteristics

Property capacity metrics (accommodates, bedrooms, beds, bathrooms) show positive associations with price, though Figure 4 reveals variance even among similar-capacity properties, suggesting that other factors (location, amenities, property type) play substantial roles in pricing.

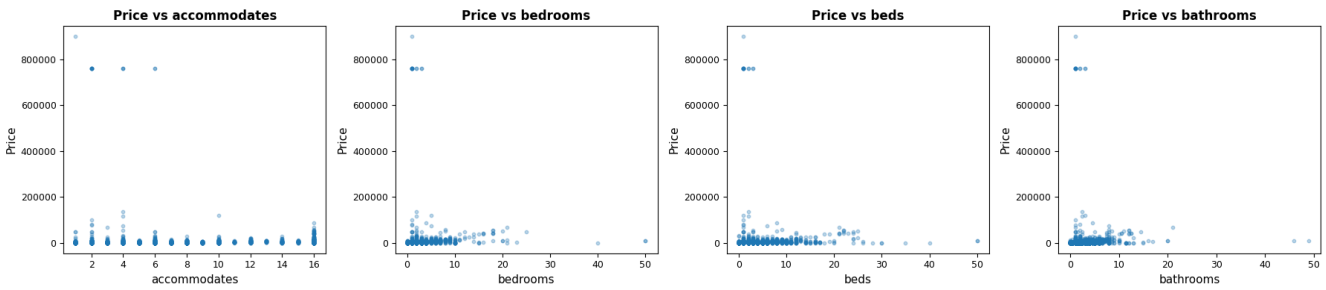

Figure 4: Scatter plots - price vs accommodates, bedrooms, beds, and bathrooms.

3.3.3 Categorical Features

Analysis of key categorical variables reveals distinct pricing patterns across accommodation types and locations.

Room Type: Table 2 shows median prices by room type, with hotel rooms highest, followed by entire homes/apartments, while shared shared rooms represent the most economical option.

Table 2: Median Price by Room Type

Room Type	Count	Median (USD)	Mean (USD)
Hotel room	63	1,994.00	88,009.73
Entire home/apt	15,570	1,276.00	1,888.26
Private room	7,289	563.00	1,508.55
Shared room	205	254.00	320.71

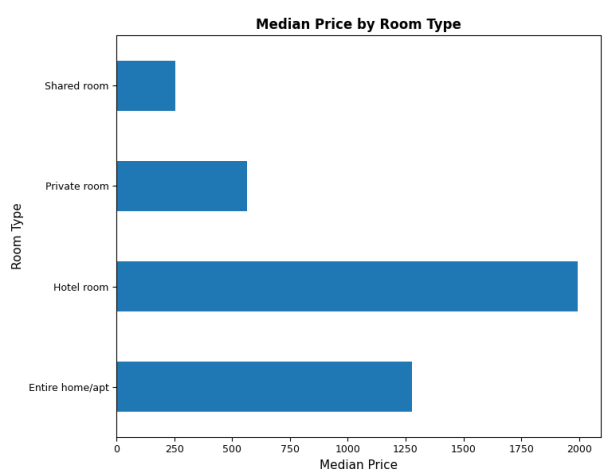

Figure 5: Bar chart - median price by room type.

Property Type: Table 3 shows "Entire rental unit" as the dominant property type in the dataset. Table 4 presents the top 10 property types by median price, revealing that unique accommodations (e.g., castles) command significantly higher prices than standard property types.

Table 3: Median Price by Dominant Property Type

Property Type	Count	Median (USD)	Mean (USD)
Entire rental unit	9,153	1,323.00	1,815.06

Table 4: Median Price by Property Type (Top 10)

Property Type	Count	Median (USD)	Mean (USD)
Castle	2	44,427.00	44,427.00
Entire villa	10	6,697.50	8,821.20
Farm stay	1	5,503.00	5,503.00
Entire home	688	2,860.00	5,624.33
Entire place	20	2,475.00	5,925.30
Entire home/apt	1	2,115.00	2,115.00
Private room in tent	1	2,037.00	2,037.00
Room in serviced apartment	13	1,990.00	2,509.46
Entire cottage	12	1,816.00	2,676.75
Entire chalet	3	1,695.00	1,350.00

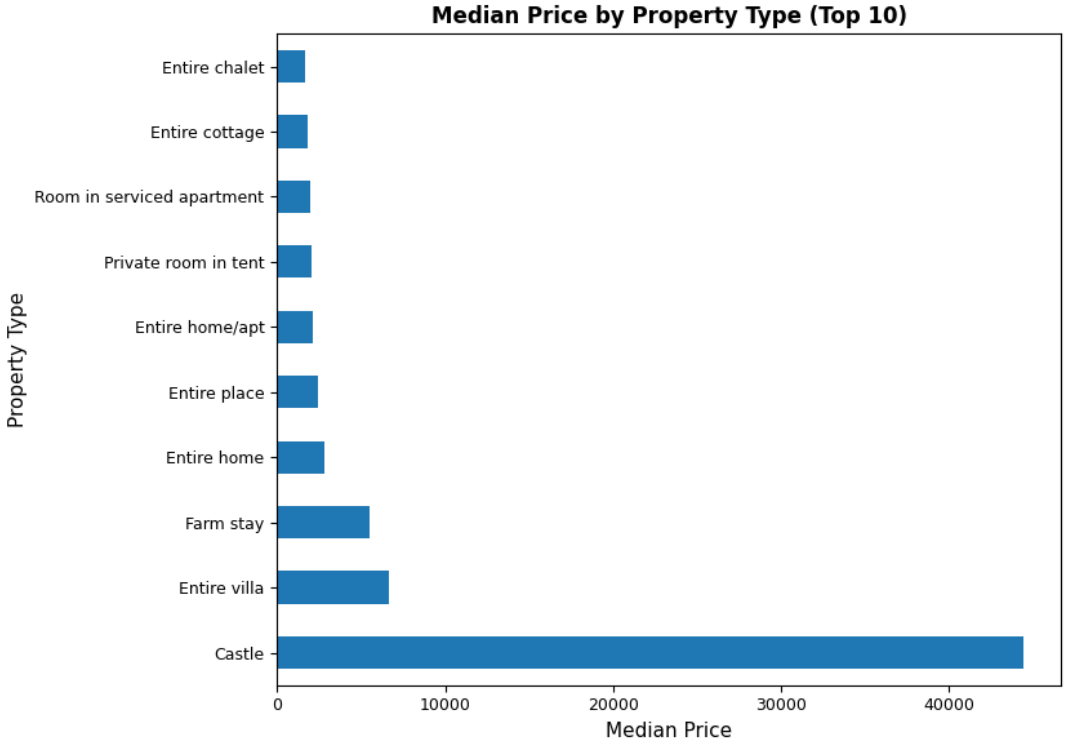

Figure 6: Horizontal bar chart - top 10 property types by median price.

Geographic Distribution: Listings are geographically concentrated in Mexico City's central areas (latitude: 19.20-19.50, longitude: -99.30 to -99.00). Figure 7 illustrates both listing density and price variation across the city, indicating that location significantly influences rental rates.

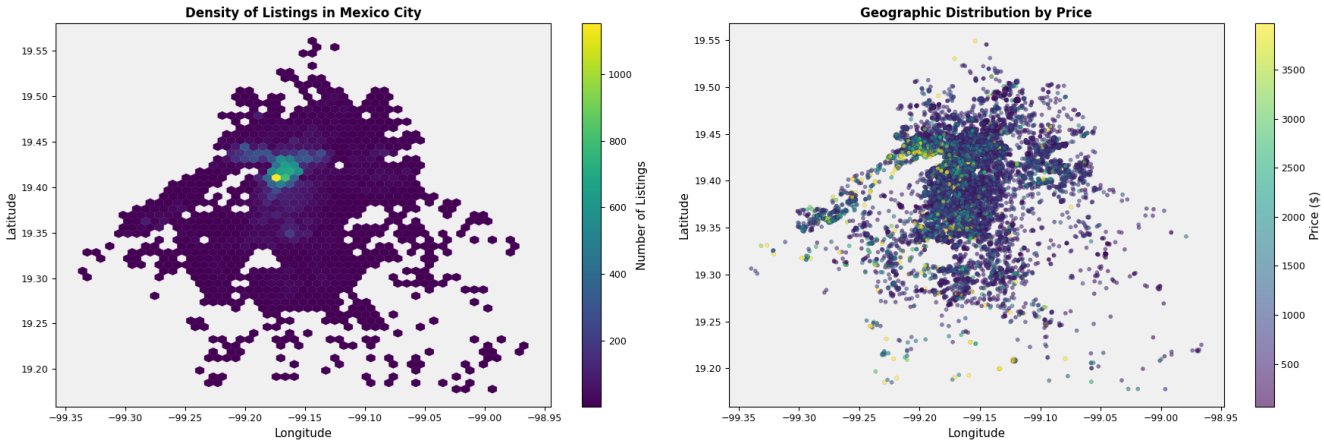

Figure 7: Geographic distribution density map and scatter plot.

References

[1] Inside Airbnb. Mexico City Airbnb detailed listings data. https://data.insideairbnb.com/mexico/df/mexico-city/2025-06-25/data/listings.csv.gz, 2025. Accessed: 2025-12.